Enhanced Classification and Segmentation of Brain Tumors in MRI Images Using Custom CNN and U-Net Models with XAI

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Abstract. This study aims to classify brain tumors in MRI images into four categories: glioma, meningioma, absent, and pituitary tumors, as well as segment low-grade gliomas. We evaluate our proposed models on four publicly available datasets to ensure robustness and generalizability. For classification tasks, we compare the performance of our custom CNN model against established models like ResNet and VGG. For segmentation tasks, we compare our custom U-Net model with the original U-Net and ResNet-based encoders. To validate the effectiveness of our models, we employ the Explainable AI (XAI) method LIME, providing insights into why our custom architectures outperform others. Our custom U-Net model achieves a validation accuracy of 99.79% and an Intersection over Union (IoU) score of 0.889 for low-grade glioma segmentation. Additionally, we report a LIME explanation stability score of 0.8169 and a sparsity score of 0.1190. The proposed custom CNN model achieves a validation accuracy of 98.70% , weighted avg precision of 97.63% , recall of 97.64% and weighted F1 - Score of 97.63%. The model achieves a LIME stability score of 0.923 and a sparsity score of 0.203.These results highlight the potential of our custom models to enhance accuracy and interpretability in brain tumor classification and segmentation tasks, offering significant improvements over existing methodologies. The custom U-net model is also an excellent negative classifier achieving a perfect 1.00 IoU score for classifying MRI scans which do not have any tumor.

Keywords: Brain Tumor · Explainable AI · LIME.

1 Introduction

Segmentation and classification of brain tumors in MRI images have been extensively researched over the past few years. Researchers have been leveraging deep neural networks to achieve significant improvements in this area [1, 2, 3, 4, 5].

However, the advent of Explainable AI (XAI) [6] has introduced new possibilities, allowing us to scrutinize network structures and understand the underlying mechanisms behind their final results. This perspective is crucial for discerning why certain models perform better than others and how existing neural network architectures can be modified for improved efficiency and accuracy.

Consequently, this can lead to smaller networks that save training time and provide faster results when deployed.
Previously, deep learning models operated largely as black boxes. Input images were fed into the network, and based on the output, additional images that led to misclassifications were incorporated into the training set to enhance learning. Developers focused on layer adjustment, feedback incorporation, loss prevention, and hyperparameter tuning, often relying on trial and error to achieve better results. With XAI, the opaque nature of neural networks has transformed into a transparent process, providing clear insights into the inner workings of the network. This transparency empowers developers to exert more control over the network, significantly reducing development time and reliance on trial and error [7].
Moreover, traditional deep learning models were constrained to predicting predefined classes without the ability to indicate uncertainty. XAI now enables us to identify when the network reaches an "I don't know" stage, allowing for the development of custom models tailored to specific needs rather than merely adapting existing network architectures for different domains through transfer learning.
Explainable AI also enhances the accountability, fairness, and transparency of deep learning models. This is particularly important in medical diagnostics, where the consequences of misclassification and segmentation can be severe.
In this paper, we develop a custom segmentation and classification model, providing interpretation with the LIME model [8]. We compare our results with benchmark neural network models such as ResNet [9] and VGG Net [10].
The paper is organized as follows: we detail our methodology in Section 2, followed by the experimentation details in Section 3 and presentation of our results and their discussion in Section4. The conclusions and implications our findings are presented in the Section 5.

2 Methodology

2.1 Proposed Neural Network for Segmentation

The proposed segmentation network is a custom U-Net, specifically designed to enhance the segmentation of brain MRI images. The architecture retains the canonical encoder-decoder structure of the original U-Net, with several enhancements aimed at capturing more complex features and improving segmentation performance. Figure 1 displays the network architecture.

Proposed Enhanced U-Net Our proposed Enhanced U-Net model introduces several critical modifications to the original U-Net architecture, aimed at augmenting its performance for brain tumor segmentation tasks. These enhancements are strategically designed to improve feature extraction, model robustness, and overall segmentation accuracy. Here are the key differences and enhancements compared to the original U-Net:

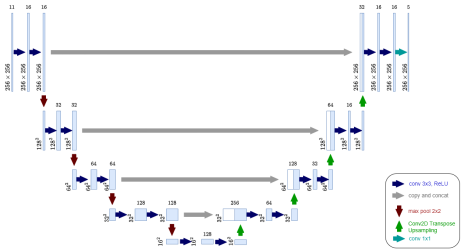

Fig. 1. Proposed Neural Network for Segmentation

Residual Connections: Residual connections are integrated within each convolutional block. This strategy helps alleviate the vanishing gradient problem and facilitates the training of deeper networks by allowing gradients to flow more effectively through the network. The residual path is implemented by adding the input to the output of the convolutional layers within the block.

Batch Normalization: Batch normalization layers are employed after each convolutional layer. This technique stabilizes and accelerates the training process by normalizing the activations, thus reducing internal covariate shift. It also allows for higher learning rates and reduces sensitivity to initialization.

Spatial Dropout: Spatial Dropout is incorporated within each convolutional block to improve generalization and prevent overfitting. This form of dropout randomly drops entire feature maps rather than individual elements, which is particularly effective for spatial data, ensuring that the model does not become overly reliant on specific feature maps.
By incorporating these enhancements, our proposed Enhanced U-Net model aims to significantly improve segmentation performance in terms of accuracy, robustness, and generalization capabilities.

2.2 Proposed Neural Network for Classification

The proposed classification network is a custom Convolutional Neural Network (CNN) specifically designed for classifying brain MRI images into benign and malignant tumors. It incorporates advanced convolutional layers with batch normalization, ReLU activations, and strategic pooling operations to enhance feature extraction and classification accuracy. Figure 2 displays the Proposed Neural Network for Classification.

Convolutional Layers: The network comprises four convolutional layers, each contributing to hierarchical feature learning:

- Conv Layer 1: Applies 32 filters of size 4x4 with a stride of 1, utilizing batch normalization and ReLU activation.

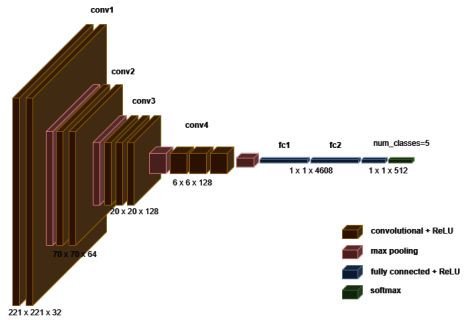

Fig. 2. Proposed Neural Network for Classification

- Conv Layer 2: Employs 64 filters of size 4x4 with a stride of 1, followed by batch normalization and ReLU activation.
- Conv Layer 3: Utilizes 128 filters of size 4x4 with a stride of 1, integrated with batch normalization and ReLU activation.
- Conv Layer 4: Applies 128 filters of size 4x4 with a stride of 1, followed by batch normalization and ReLU activation.

Pooling Layers: Pooling operations are strategically placed to reduce spatial dimensions and enhance translational invariance:

- Pooling Layer 1: After the first and second convolutional layers, a max pooling operation with a kernel size of 3x3 and a stride of 3 is applied.
- Pooling Layer 2: Following the third convolutional layer, a max pooling operation with a kernel size of 3x3 and a stride of 2 is employed.

Fully Connected Layers: After feature extraction through convolution and pooling, the network incorporates fully connected layers for final classification:

- Fully Connected (FC) Layer 1: Composed of 512 units with ReLU activation, facilitating complex feature integration.
- Dropout Regularization: Implemented with a dropout rate of 0.5 before the final classification layer to prevent overfitting by randomly deactivating neurons during training.
- Final Classification FC Layer 2: The last layer outputs predictions corresponding to the number of classes, leveraging integrated features for accurate classification.

Enhanced Performance: The strategic use of two different strides in the max pooling layers significantly enhances test accuracy, allowing the network to effectively capture and integrate hierarchical features from varying spatial scales. This design choice optimizes the network's ability to discern between benign and malignant brain tumor images with increased precision and reliability in medical imaging applications.

2.3 Applying LIME on Proposed Models

Application on Classification Models: For the classification task, LIME is applied to understand the reasons behind the classification of brain MRI images as benign or malignant or no tumor. The steps include:

1. Selecting Instances: A subset of correctly classified and misclassified images from the test set is selected for explanation.
2. Generating Explanations: LIME generates explanations for each selected instance by highlighting the regions of the image that most influenced the model's decision.
3. Visualizing Explanations: The explanations are visualized as heatmaps overlaid on the original images, showing which parts of the image contributed most to the classification.

Application on Segmentation Models: For the segmentation task, LIME helps to explain why certain regions of the MRI images were segmented as tumor areas. The steps include:

1. Selecting Instances: A set of segmented images, including both successful and failed segmentations, is chosen for explanation.
2. Generating Perturbations: Perturbations are applied to the images, and the model's segmentation predictions for these perturbed samples are obtained.
3. Fitting an Interpretable Model: A simpler model is fitted to approximate the segmentation decisions of the original model.
4. Visualizing Explanations: The important features (image segments) that influenced the segmentation decision are visualized, helping to understand the model's behavior.

2.4 Algorithm for the Proposed Work

The following steps outline the algorithm we employed in our research for preprocessing data, training custom models for classification and segmentation, applying LIME for explainability, and evaluating the models using specific metrics.

Step 1: Data Preparation:

1. Data Acquisition: We collected a publicly available dataset of brain MRI images. The dataset includes labels for classification (glioma , pituitary , meningioma ,no tumor) and ground truth masks for segmentation.
2. Data Preprocessing:We normalized the images to a standard scale (e.g., [0, 1]). The images were resized to a fixed size (e.g., 256x256 pixels) to ensure uniform input dimensions. Used a custom data loading function to load , preprocess MRI Images and load them in a Dataframe. We augmented the dataset using techniques such as cropping out the brain sections only to enhance model generalization and accuracy and also adding data augmentations such as horizontal flips and rotations.

Step 2: Model Training:

1. Classification Model: We defined a custom CNN architecture for classification. The model parameters were initialized, and the model was compiled with an appropriate optimizer (Adam) and loss function (cross-entropy loss). The dataset was split into training, validation, and test sets. We trained the model on the training set and validated it on the validation set. The training process was monitored, and early stopping was applied to prevent overfitting.
2. Segmentation Model: We defined a custom U-Net architecture for segmentation. The model parameters were initialized, and the model was compiled with an appropriate optimizer (Adam) and loss function (binary cross-entropy with Dice coefficient). The model was trained on the training set and validated on the validation set. The training process was monitored, and early stopping was applied to prevent overfitting.

Step 3: Model Evaluation

1. Classification Evaluation: We evaluated the trained classification model on the test set using metrics such as accuracy, precision, recall, and F1 score.
2. Segmentation Evaluation: We evaluated the trained segmentation model on the test set using metrics such as Dice coefficient, Intersection over Union (IoU), and pixel-wise accuracy.

Step 4: Applying LIME for Explainability

1. Instance Selection: We selected a subset of correctly classified and misclassified images from the test set for classification explainability. We selected a set of successful and failed segmentations from the test set for segmentation explainability.
2. Generate Explanations: We applied LIME to generate explanations for the selected instances. For classification, we created perturbed samples and fit an interpretable model to approximate the classifier's behavior locally. For segmentation, we created perturbed samples and fit an interpretable model to approximate the segmenter's behavior locally.
3. Visualization and Interpretation: We visualized the explanations as heatmaps to highlight the regions of the images that contributed most to the model's predictions. The visualizations were interpreted to understand the model's decision-making process.

Step 5: Quantitative Evaluation of Explanations

1. Explanation Stability: We measured the consistency of the explanations when the input image was slightly perturbed.
2. Explanation Sparsity: We evaluated the proportion of the image highlighted in the explanation to assess conciseness.
3. Explanation Fidelity: We assessed how well the interpretable model approximated the original model's predictions.

3 Experimental Details

3.1 Dataset Used

For Classification Tasks: For the classification tasks, we employed a combined dataset comprising 7023 images of human brain MRI images. These images are categorized into four distinct classes: glioma, meningioma, no tumor, and pituitary. This dataset amalgamates images from multiple sources, providing a diverse and comprehensive collection for robust classification model training and evaluation.

Sources of Data:

1. Jun Cheng's Brain Tumor Dataset [11, 12]: This dataset contains 3064 T1-weighted contrast-enhanced images from 233 patients, distributed across three tumor types: meningioma (708 slices), glioma (1426 slices), and pituitary tumor (930 slices). The dataset is split into four subsets, each archived in a .zip file containing approximately 766 slices. The data is organized in MATLAB (.mat) format, with each file storing the image data and associated annotations.
2. Br35H Dataset: The Br35H dataset contains a diverse collection of brain MRI images used for various brain tumor classification tasks. The dataset includes images categorized into the four classes mentioned above, further enhancing the diversity and robustness of our classification model.
3. SarTaj Dataset: This dataset includes additional brain MRI images used to complement the classification model's training dataset, ensuring a robust learning process.

The combined dataset from these sources provided a rich variety of images, enhancing the model's ability to generalize across different types of brain tumors. The images in the combined dataset were preprocessed to ensure uniformity in resolution and intensity normalization, followed by augmentation techniques to further increase the dataset size and variability, such as rotation, flipping, and contrast adjustment.

For Segmentation Tasks:

Brain MRI Segmentation Dataset: This dataset includes brain MR images along with manual FLAIR abnormality segmentation masks. The images were sourced from The Cancer Imaging Archive (TCIA) and correspond to 110 patients included in The Cancer Genome Atlas (TCGA) lower-grade glioma collection, each having at least one fluid-attenuated inversion recovery (FLAIR) sequence and associated genomic cluster data. FLAIR sequences, known for their high sensitivity to lesions and abnormalities within the brain tissue. Manual segmentation masks were created by expert radiologists, delineating the abnormal regions with high precision. Covers 110 patients, providing a comprehensive and diverse dataset

for training and evaluating segmentation models.
Each image has been standardized to a uniform resolution, ensuring consistency across the dataset. Images were preprocessed to correct for intensity inhomogeneities and were normalized to have zero mean and unit variance. Data augmentation techniques, including random rotations, scaling, and elastic deformations, were applied to increase the effective size of the training set and to improve the robustness of the segmentation model. The segmentation masks provide a binary representation of the tumor regions, with 1s indicating the presence of a tumor and 0s representing healthy tissue. These masks are crucial for training supervised learning models for segmentation tasks.

3.2 Evaluation Metrics

Segmentation : For evaluating the performance of our segmentation models, we employed several standard metrics to ensure a comprehensive assessment:

1. Dice Coefficient (Dice Similarity Index, DSC):The Dice coefficient measures the overlap between the predicted segmentation and the ground truth. It ranges from 0 to 1, with 1 indicating perfect overlap.

$$D(A, B) = \frac{2\,|A \cap B|}{|A| + |B|} \tag{1}$$

   where A is the set of pixels in the predicted segmentation and B is the set of pixels in the ground truth segmentation.
2. Intersection over Union (IoU, Jaccard Index):IoU measures the ratio of the intersection to the union of the predicted and ground truth segmentations. It ranges from 0 to 1, with 1 indicating perfect segmentation.

$$\text{IoU} = \frac{\text{Area of overlap}}{\text{Area of union}} = \frac{A \cap B}{A \cup B} \tag{2}$$

3. Precision (Positive Predictive Value): Precision indicates the proportion of true positive pixels among all pixels that were predicted as positive.

$$\text{Precision} = \frac{\text{TP}}{\text{TP+FP}} \tag{3}$$

   where TP is the number of true positive pixels and FP is the number of false positive pixels.
4. Recall (Sensitivity, True Positive Rate): Recall measures the proportion of true positive pixels among all actual positive pixels.

$$\text{Recall} = \frac{\text{TP}}{\text{TP+FN}} \tag{4}$$

   where FN is the number of false negative pixels.

5. F1 Score: The F1 score is the harmonic mean of precision and recall, providing a single metric that balances both.

$$\text{F1 Score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{5}$$

6. Hausdorff Distance: This metric measures the maximum distance between the predicted segmentation boundary and the ground truth boundary, providing insight into the spatial accuracy of the segmentation.

$$d_H\left(X, Y\right) = \max\left\{\sup_{x \in X} d\left(x, Y\right), \sup_{y \in Y} d\left(X, y\right)\right\} \tag{6}$$

where sup represents the supremum operator and $d(a, B) = \inf_{b \in B} d(a, b)$. inf is the infimum operator $d(a, B)$ quantifies the distance from a point $a \in X$ to the subset $B \subseteq X$. $d(a, b)$ is the Euclidean distance between points a and b.

Classification : For the classification tasks, the following metrics were utilized to evaluate the model performance:

1. Accuracy: Accuracy is the ratio of correctly predicted instances to the total instances. It provides a straightforward measure of overall performance.
2. Confusion Matrix: The confusion matrix provides a detailed breakdown of the classification performance, displaying the counts of true positives, true negatives, false positives, and false negatives for each class.
3. Receiver Operating Characteristic (ROC) Curve and Area Under the Curve (AUC): The ROC curve plots the true positive rate against the false positive rate at various threshold settings. The AUC provides a single scalar value summarizing the model's performance across all thresholds. An AUC of 1 indicates perfect classification, while an AUC of 0.5 suggests no better performance than random guessing.

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \tag{7}$$

Precision, Recall and F1-scores are also used to evaluate the results.

LIME (Local Interpretable Model-Agnostic Explanations) : For the interpretability of our models, particularly in understanding their decision-making processes, we employed LIME to generate explanations. The following metrics were used to evaluate the quality of the explanations provided by LIME:

1. Explanation Stability: Explanation stability measures the consistency of explanations when slight perturbations are made to the input data. High stability indicates that small changes in the input do not significantly alter the explanation.

$$\text{Stability} = 1 - \frac{1}{n} \sum_{i=1} n \left|E\left(x_i\right) - E\left(x_i'\right)\right| \tag{8}$$

where $E\left(x_i\right)$ is the explanation for instance x_i and $E\left(x_i'\right)$ is the explanation for the perturbed instance and n is the number of samples.

2. Explanation Sparsity: Explanation sparsity evaluates the proportion of features used in the explanation compared to the total number of features. Sparse explanations are preferred as they are easier to interpret.

$$\text{Sparsity} = 1 - \frac{\text{Number of features in explanation}}{\text{Total number of features}} \tag{9}$$

3. Explanation Fidelity: Explanation fidelity measures how well the explanation approximates the original model's behavior. High fidelity indicates that the surrogate model used for generating explanations closely mimics the original model.

$$\text{Fidelity} = 1 - \frac{1}{n}\sum_{i=1}^{n}\left(f\left(x_i\right) - g\left(x_i\right)\right)^2 \tag{10}$$

where $f\left(x_i\right)$ is the prediction of the original model for instance x_i and $g\left(x_i\right)$ is the prediction of the surrogate model.

3.3 Setting Up Parameter Values

For both the segmentation and classification tasks, careful selection and tuning of hyperparameters were crucial to optimizing the performance of our neural network models. The parameter values for various components of our proposed work are detailed below:

Segmentation Model (Enhanced U-Net)

1. Learning Rate: 1e-4. The learning rate determines the step size at each iteration while moving toward a minimum of the loss function. A smaller learning rate ensures a stable convergence but might require more epochs.
2. Batch Size: 16. The batch size indicates the number of training samples used in one forward/backward pass. A moderate batch size balances memory efficiency and gradient stability.
3. Epochs: 50 The number of epochs defines how many times the learning algorithm will work through the entire training dataset. More epochs can lead to better convergence but may also increase the risk of overfitting.
4. Dropout Rate:
   - Initial layers: 0.1
   - Middle layers: 0.2
   - Final layer: 0.3

   Dropout is used to prevent overfitting by randomly setting a fraction of input units to 0 at each update during training time. Different rates are used for different layers to balance regularization and learning.
5. Optimizer Adam. The Adam optimizer is chosen for its adaptive learning rate capabilities and efficient handling of sparse gradients, making it suitable for training deep neural networks.

6. Loss Function: Binary cross entropy is used for measuring the performance of a classification model whose output is a probability value between 0 and 1. It is well-suited for segmentation tasks where the output is a binary mask.

$$\text{Binary Cross Entropy} = -\frac{1}{N}\sum_{i}^{N}\sum_{j}^{M} y_{ij}\log\left(p_{ij}\right) \tag{11}$$

where N is the number of samples and M is the number of classes.

Classification Model (Custom CNN)

1. Learning Rate: 1e-3. A higher learning rate compared to the segmentation model to ensure faster convergence while maintaining stability.
2. Batch Size: 32. A larger batch size to improve gradient estimation accuracy and training speed.
3. Epochs: 100. A higher number of epochs to ensure sufficient training for convergence.
4. Dropout Rate: 0.5. A higher dropout rate to strongly regularize the model and prevent overfitting, given the smaller dataset size.
5. Optimizer Adam. Adam optimizer is chosen for its efficient gradient computation and adaptive learning rates, facilitating robust training.
6. Loss Function: Cross Entropy Loss. Cross entropy loss is used for multi-class classification tasks, where it evaluates the performance of a classification model whose output is a probability value between 0 and 1 for each class.

$$\text{Cross Entropy} = -\frac{1}{N}\sum_{j=1}^{N}\left[t_j\log\left(p_j\right) + \left(1-t_j\right)\log\left(1-p_j\right)\right] \tag{12}$$

N is the number of data points, t_j is the truth value and p_j is the Softmax probability for taking the truth value.

LIME Parameters

1. Number of Samples: 1000. The number of perturbed samples generated to explain each prediction. More samples can improve explanation fidelity but increase computational cost.
2. Kernel Width: 0.25. The width of the kernel used for weighting the perturbed samples. A smaller width focuses the explanation on samples closer to the original instance.
3. Feature Selection Method: Forward Selection. The method used for selecting the most important features in the perturbed samples. Forward selection iteratively adds features to improve explanation quality.
4. Regularization: L1 Regularization. L1 regularization encourages sparsity in the explanation, making it more interpretable by focusing on the most influential features.

5. Segmenter: Quickshift
   - Kernel Size: 4. Controls the spatial scale of the segmentation. A larger kernel size results in larger segments.
   - Max Distance: 200. Limits the distance in the color space between two pixels to be merged.
   - Ratio: 0.5. Balances the color proximity and spatial proximity. A higher ratio gives more importance to color similarity.

   Quickshift is a mode-seeking segmentation algorithm that clusters pixels based on color similarity and spatial proximity. It is used to generate superpixels, which are smaller segments of the image that preserve local information.

4 Results and Discussion

Table 1 presents the evaluation results of our proposed Custom CNN model compared to ResNet32 and VGG16 models.

The Custom CNN model achieved the highest accuracy (98.70%) compared to ResNet32 (96.35%) and VGG16 (96.2%), indicating superior overall performance in classifying brain tumors. The Custom CNN model also demonstrated the highest precision (97.63%), compared to ResNet32 (95.24%) and VGG16 (95.78%), indicating a lower rate of false positives. Recall: The Custom CNN model achieved a recall of 97.64%, slightly higher than ResNet32 (96.12%) and VGG16 (95.12%), indicating a higher rate of true positives. The Custom CNN model showed the highest F1-Score (97.47%), compared to ResNet32 (96.15%) and VGG16 (95.76%), balancing precision and recall effectively. The Custom CNN model had the highest LIME Explanation Stability Score (0.923), compared to ResNet32 (0.846) and VGG16 (0.687), indicating more stable explanations across similar inputs. The Custom CNN model had a LIME Explanation Sparsity Score of 0.208, compared to ResNet32 (0.196) and VGG16 (0.225), with lower sparsity indicating more concise explanations. ResNet32 had the highest LIME Explanation Fidelity Score (0.556), followed by VGG16 (0.418), and the Custom CNN model (0.310). This measures how well the explanation model approximates the original model.

These results demonstrate that the Custom CNN model generally outperforms both ResNet32 and VGG16 in terms of classification accuracy, precision, recall, and F1-Score, while providing highly stable and reasonably concise explanations as measured by LIME.

Table 2 presents the evaluation results of our proposed Custom U-Net model compared with a U-Net model that uses ResNet as its encoder. The Custom U-Net model achieved a higher validation accuracy (99.8%) compared to the U-Net with ResNet as its encoder (99.11%). The Custom U-Net model demonstrated a lower validation loss (0.0132) than the U-Net with ResNet (0.0425), indicating better performance in minimizing error. The IoU score, which measures the overlap between the predicted and ground truth segments, was higher for the Custom U-Net model (0.889) compared to the ResNet-based U-Net (0.847), suggesting more accurate segmentation. The Custom U-Net model had a higher LIME

Table 1. Evaluation Results of proposed Custom CNN model compared to ResNet and VGG16

Evaluation Metrics	**Custom CNN Model**	**ResNet32**	**VGG16**
Accuracy	98.70%	96.35%	96.2%
Precision	97.63	95.24	95.78
Recall	97.64	96.12	95.12
F1-Score	97.47	96.15	95.76
LIME Explaination Stability Score	0.923	0.846	0.687
LIME Explaination Sparsity Score	0.208	0.196	0.225
LIME Explaination Fidelity Score	0.310	0.556	0.418

Explanation Stability Score (0.8169) versus the ResNet-based U-Net (0.7873), indicating more stable explanations across similar inputs. Both models showed similar LIME Explanation Sparsity Scores, with the Custom U-Net model having a slightly lower score (0.1190) compared to the ResNet-based U-Net (0.1221). Lower sparsity indicates more concise explanations. The ResNet-based U-Net model exhibited a higher LIME Explanation Fidelity Score (0.6036) compared to the Custom U-Net model (0.5447), which measures how well the explanation model approximates the original model. Figure 3 displays the Confusion Matrix

Table 2. Evaluation Results for the proposed Custom U-net model compared with ResUnet

Evaluation Metrics	**Custom U-net Model**	**U-net with ResNet as Encoder**
Validation Accuracy	99.8%	99.11%
Validation Loss	0.0132	0.0425
Intersection over Union Score (IoU)	0.889	0.847
LIME Explaination Stability Score	0.8169	0.7873
LIME Explaination Sparsity Score	0.1190	0.1221
LIME Explaination Fidelity Score	0.5447	0.6036

of custom CNN. Figure 4 displays the Plot of ground truth scans , masks and predicted masks and labels along with IoU Score. Figure 5 displays the MRI Scan on which LIME visualizations are generated. Figure 6 displays the LIME Visualizations of Custom CNN with feature heatmap , positive and negative influences and top three feature visualization. Figure 7 displays the plot of ground truth scans , masks and predicted masks and labels along with IoU Score Negative Classifier acheiving perfect IoU of 1.

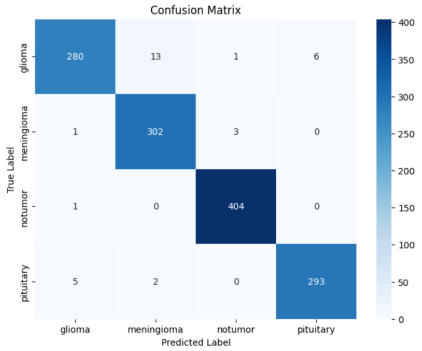

Fig. 3. Confusion Matrix of custom CNN

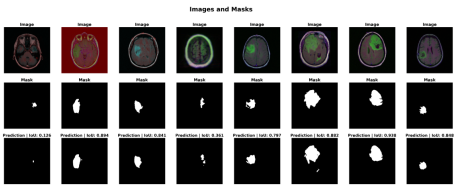

Fig. 4. Plot of ground truth scans , masks and predicted masks and labels along with IoU Score

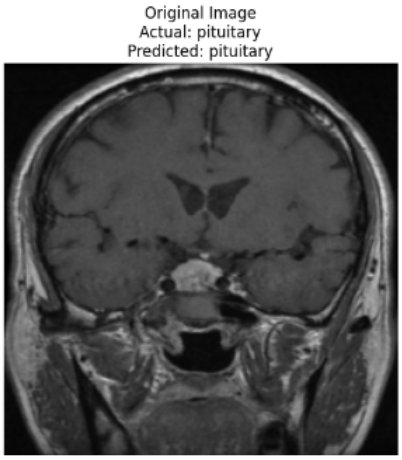

Fig. 5. MRI Scan on which LIME visualizations are generated

5 Conclusion and Future Scope

In this study, we developed and evaluated advanced neural network architectures for brain tumor classification and segmentation using MRI images. Our enhanced

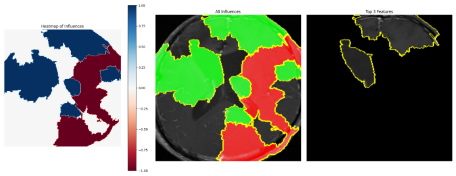

Fig. 6. LIME Visualizations of Custom CNN with feature heatmap , positive and negative influences and top three feature visualization

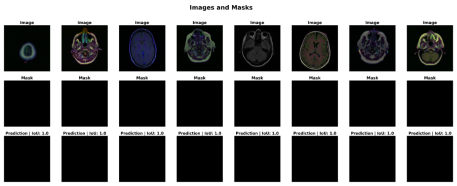

Fig. 7. Plot of ground truth scans , masks and predicted masks and labels along with IoU Score Negative Classifier acheiving perfect IoU of 1

U-Net model demonstrated improved segmentation performance, while our custom CNN showed significant accuracy in classification tasks. The integration of Explainable AI (XAI) techniques, particularly LIME, provided valuable insights into the model's decision-making processes, enhancing interpretability and trustworthiness.

Combining multiple neural network architectures, such as integrating attention mechanisms or transformers, could improve both classification and segmentation performance.Leveraging larger and more diverse datasets can improve the generalizability of our models across different populations and imaging modalities. Developing models capable of real-time processing can facilitate clinical applications, enabling quicker diagnosis and treatment planning. Incorporating more advanced XAI techniques can provide deeper insights into model behavior, improving interpretability and clinical acceptance. Utilizing pre-trained models on broader datasets and fine-tuning them on specific medical imaging tasks can enhance model performance and reduce training time. Implementing these models in clinical workflows and electronic health records (EHR) systems can aid in automated diagnosis and decision support.

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